

Software Requirements and Design (SRD) Report

Project Title: PixelPals
Module: CS2TP |Team: 23 | Date: 14.11.2025

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1. Team Information

1.1 Team Roles and Statement of Contribution

The PixelPals team contributed to the planning and documentation of the software requirements and design report (SRD) in Term 1A, while discussing the plan for Term 1B, which focuses on implementing the functional website. Each member's role, Belbin role, contributions to Term 1A and plan for Term 1B are summarised below.

Table 1.1.1: Summary of Team Roles and Contributions

Maeesa Ali: Project manager & back-end support (Coordinator)
Statement of Contribution for Term 1A: Completed report sections 1.1,2.4,4.2.2,6.1,6.2,6.4 and appendix F. Organised Trello board, GitHub structure and all folders, team roles and task allocation, maintained repository merges and overall document structure.
Planned Contribution for Term 1B: Back-end support, including authentication workflow, parental-control logic, database setup and coordination of merges and testing.
Toney Shajan: Quality Assurance (Monitor-Evaluator)
Statement of Contribution for Term 1A: Completed Sections 1.1, 4.3, 6.3, 7.0 and reviewed SRD drafts for accuracy and consistency.
Planned Contribution for Term 1B: Usability, accessibility, and navigation testing across devices and user flows.
Russell Chamabe: Back-end Developer (Implementer)
Statement of Contribution for Term 1A: Completed Sections 2.4, 3.1, 5.4, and Appendix A. Contributed to requirements discussions and supported backend planning decisions.
Planned Contribution for Term 1B: Back-end product-management logic, database interactions, and Laravel model/controllers.
Joel Gyan Baffour: Systems Tester & Back-end Support (Completer-Finisher)
Statement of Contribution for Term 1A: Completed Sections 3.1.1, 4.1, 4.2.1, 1.5, 5.5, and Appendix B. Researched technical security topics and contributed to requirement accuracy.
Planned Contribution for Term 1B: Systems testing (validation, error handling) and back-end data validation.
Seher Yaser: UI/UX Designer (Plant)
Statement of Contribution for Term 1A: Completed Sections 1.2, 4.3, 5.3, 5.6 and Appendices C & E. Created the team logo and contributed to layout, accessibility and UI decisions.
Planned Contribution for Term 1B: Wireframes, colour schemes, high-fidelity UI design, and page styling.
Oscar Metcalf: Front-end Developer (Resource Investigator)

Statement of Contribution for Term 1A: Completed Sections 3.1.1, 5.1 and Appendix E. Contributed research on functional requirements and provided design input.
Planned Contribution for Term 1B: Front-end product page development (layout, filtering, interactions).
Jamaal Pickford: Front-end Developer (Shaper)
Statement of Contribution for Term 1A: Completed Sections 3.1.1, 5.2, 5.5 and Appendix D. Created a user-journey diagram and contributed to requirement drafting.
Planned Contribution for Term 1B: Front-end login and basket interfaces, error states, and user-navigation behaviour.
Dia Kaur: Content Creator (Team worker)
Statement of Contribution for Term 1A: Completed Sections 2.1–2.3, 6.3, and researched audience needs and tone-of-voice preferences.
Planned Contribution for Term 1B: Written UI content, product descriptions, and accessible alt-text for images.

1.2 Team Logo and Identity

PixelPals is an exciting, playful, and tech-driven e-commerce brand which creates fun and ergonomic gaming accessories for kids. The logo displayed in diagram 1.2.1 reflects the brand's identity through a retro, pixel art style, creating a nostalgic connection for parents while also engaging younger, modern audiences through its vibrant and lively colours.

Diagram 1.2.1: PixelPals Logo



2. Project Description

2.1 Project Overview

PixelPals is a website aimed at promoting appropriate technology to parents and children. The initiative addresses concerns over physical and cognitive effects of prolonged screen time-posture, hearing, vision, and fine motor skills (Elghomati et al., 2022). PixelPals offer a range of ergonomically designed accessories, including adaptive keyboards and mice, supportive seating, volume-sensitive headphones, and screen-time charts.

2.2 Vision and Objectives

The aim is to reduce physical strain and stunting of motor skills, educate families on safe and sustainable device habits and promote inclusive, user-centred designs aligned with developmental needs.

2.3 Target Audience

PixelPals' target audience is parents and children aged 3 to 12. But due to the focus on ergonomics, PixelPals can support elderly and disabled users, making the design versatile and expansive.

2.4 Technologies and Tools

PixelPals is going to use certain technologies and tools for the current design stage and for future development of the system, which is displayed in Table 2.4.1

Table 2.4.1: Planned Technologies and Their Purpose

Category	Technology and Tools used/planned	Purpose
Backend development	Laravel (PHP), MySQL	Will be used to make user accounts, product data and parental controls.
Frontend development	HTML, CSS, JavaScript	Will be used to make the website interface and for browsing and selecting products.
UI/UX design	Figma, Canva	Used to create layout, logo and accessibility structure.
Version Control	GitHub	Used to store project work and control changes by using pull requests and commits.
Project Management	Trello, Discord	Trello is used to assign and track tasks with deadlines. Discord is used for team communication.
Documentation	Microsoft Word	Used to make and update the Software Requirements Design Report.

3. Functional Requirements

3.1 Introduction

Functional Requirements are based on what the system should do. This can be done by the usage of features, specific actions, and behaviours. Furthermore, it will make sure that the system allows the users to register, sign in, place orders and add parental control.

Table 3.1.1: Functional Requirement (FR) List

FR1: Product Search:
Users can search for products by entering keywords in a search bar. Products with names or descriptions matching the keywords are displayed. If no matches are found, a message appears stating that no products match the search.
FR2: Product Filter:
Users can refine their product search using filters such as price and age range. Multiple filters can be applied simultaneously, and results update instantly. Filters can also be reset to show all products.
FR3: Login / Registration:
When accessing the website, users are prompted to either log in or register. During registration, the user's details (e.g. username and password) are gathered and stored in a relational database. During login, the user's given details are checked and verified against the stored information to validate the login.
FR4: Basket / Checkout:
Users can select products, which would then copy and send their data to an array which would act as a basket for the user. Once finished, the user can input their payment details at checkout to finalise their purchase.
FR5: Account Management:
The system should allow the user to change personal information such as phone number, email address, home address, etc. The system will override the previous information and update the changes made by the user.

4. Non-Functional Requirements

4.1 Performance

The website must be able to load quickly, as it'll prevent users from leaving the website. To boost perceived performance, there should be immediate visual feedback, for example, a button changing colour when clicked or showing a loading icon so that they are aware that their action has been registered and that the website is processing it.

4.2 Security

4.2.1 Technical Security

To protect user data, PixelPals will implement a Hypertext Transfer Protocol Secure (HTTPS) by using a valid TLS certificate to encrypt data sent between the browser and the website to prevent man-in-the-middle attacks.

Furthermore, Passwords must have a minimum of 8 characters, and they must include an uppercase letter, a lowercase letter and a number or special character. Old passwords can't be reused, and passwords can't be shared, and if someone finds out the password, it must be changed.

4.2.2 Parental Control and Data Privacy

The PixelPals website is designed for children and parents, which means data protection must be prioritised at every stage of development. To achieve this, the system will include parental control features and account security to support safe and responsible use. The security requirements and how they will be implemented are stated below:

- **Child data privacy:** All information will be securely stored in the database to prevent unauthorised access, and only essential data will be collected from users.
- **Parental control access:** Parents will have access to screen time rules for children.
- **Password security:** Passwords will be hidden on screen during login, and the system will prevent children from giving their passwords to others.
- **Access permissions:** Only authorised team members will be able to modify or update sensitive product or system information.

Overall, these measures will keep the system suitable for children while giving parents reassurance and control over their child's digital wellbeing.

4.3 Usability and Accessibility

The PixelPals website is designed to be easy to use for children and parents. It follows key UX principles, which include clear navigation, hierarchy and accessibility. Pages have a consistent theme, with high contrast and readable fonts, good for those with cognitive needs.

The interface will be tested on different devices to make sure it functions. Accessibility tools like WAVE and axe DevTools will screen-reader compatibility, along with manual checks to confirm keyboard navigation. Each step will make sure the website meets WCAG 2.1 accessibility standards.

4.4 Maintainability

The PixelPals system is designed to be easy to update and expand in the future. Table 4.4.1 shows how the team has used organised and traceable development practices to support all continuous improvements.

Table 4.4.1: Maintainability principles and their application in PixelPals

Maintainability principle	Application in PixelPals
Clear folder structure	Creating organised files that are easy to locate, such as SRD, Drafts, Evidence and Docs files on GitHub.
Version control best practice	Team members will be their own branches on GitHub and use pull requests to the main area to avoid conflicts and track changes early.
Continuous documentation	All meeting notes, commit summaries and report document versions will be recorded throughout development.
Modular design approach	Frontend and Backend components will be built in small functional parts for easier updates and fixes.

To summarise, all these approaches will support long-term maintenance by allowing new features and bug fixes to be added without difficulty.

5. Use Cases and Design

5.1 Use Cases-Products

The product use case diagrams show how users interact with the catalogue by viewing product details by searching, and filtering. The use case diagrams are linked to the product search and filtering functional requirements, which are mentioned and described in section 3.2.

Diagram 5.1.1: Product use case diagram (search and view).

Use case number: 1	Name: Search products
Goal: To allow users (customers) to search for gaming accessories using keywords, making it easy to find specific products.	
Actors: User (customer)	
Frequency of execution: High users often search for products to make searching for specific accessories more efficient	
Scalability: The system should be able to handle several searches enquires at once which will be very important in peak times	
Criticality: Medium, searching helps the experience more user friendly and efficient for the customer but not needed for the system to work, as they can still find the accessories manually	
Other non-functional requirements: search results should load quickly, results should be case insensitive and match partial keywords, the design must be child friendly, simple design with clear results	
Preconditions: Website is open, products exist, the user is on the products or home page	
Postconditions: products matching users search are on the screen, the user can browse and refine their search	
Primary path: user clicks on search bar, user types keywords, the system checks for products tiles and descriptions that match the keywords, the products appear in the product grid, user can scroll through the grid and select a product	
Use cases related to primary path: filter products (use case 2) user may want to refine the results after searching	
Alternatives: The user types nothing in the search bar, the user presses enter, the entire product catalogue appears The user types in a generic term such as "gaming" and a broad range of products shows up	
Use cases related to alternatives: View all products	
Exceptions: the user types a keyword that doesn't match any product, the page says "no products found"	
Use cases related to exceptions: filter product (to narrow down results), view all products (reset search)	

Diagram 5.1.2: Product use case diagram (filtering and selection).

Use case number: 2	Name: Filter products
Goal: To allow users (customers) to filter gaming accessories from the selected categories using ranges such as price, age, popularity, making it easier to find products matching the users desired requirements	
Actors: User (customer)	
Frequency of execution: High users often want to narrow down their searches further to match their specific needs	
Scalability: The system should be able to update results in real time even when there is a lot of products to filter	
Criticality: Medium, filtering helps increase the satisfaction for the customer but it isent needed for the system to work as they can still find the specific accessories manually	
Other non-functional requirements: filtered results should load quickly, results should match the range selected by the user, the filters should be clear and simple (price, age, popularity), users should be able to use multiple filters at once	
Preconditions: Website is open, products are loaded, the products are currently displayed	
Postconditions: products update to match the filters criteria, filters remain visible and adjustable by the users	
Primary path: user open sidebar or menu on the products page, user selects one or more filters, the system refreshes for the products the products on the page to match the filters, the filters can be added and removed in order to instantly change the products displayed	
Use cases related to primary path: Search products (use case 2) user may want to search for specific products that have these filters View products details as the user may want to see more info on the product	
Alternatives: The user clears all filters, so the system resets to show all products User chooses filters that contradict, and the system shows no results	
Use cases related to alternatives: search products as the user might want to perform a new search instead of filtering	
Exceptions: the user chooses filters that does not match any product, the page says "no products found"	
Use cases related to exceptions: view all products (to show the results of the filter reset)	

5.2 Use Cases – Login & Basket

Use case diagrams linking to login and basket show how users should sign in, sort out their basket and checkout. The diagrams below support the functional requirements for login, basket and account management.

Diagram 5.2.1: Login use case overview.

Use Case: Account management
Goal: To allow users to manage their account details
Actor(s): User
<u>Preconditions:</u> • The user is logged in
<u>Main Flow:</u> 1. The user navigates to the “manage account” page 2. The system displays the user’s stored information 3. The user selects and updates one or more fields 4. The system validates the changes 5. The system commits the changes to its database 6. A confirmation message is displayed
<u>Post conditions:</u> • The updated user information is stored in the relational database
<u>Alternate flows:</u> • Invalid input: If invalid data is input, an error message is displayed, and the user is prompted to try again • Database error: If the update fails, the system logs the error, and the user is prompted to try again

Diagram 5.2.2: Basket management use case.

Use Case: Account management
Goal: To allow users to manage their account details
Actor(s): User
<u>Preconditions:</u> • The user is logged in
<u>Main Flow:</u> 1. The user navigates to the “manage account” page 2. The system displays the user’s stored information 3. The user selects and updates one or more fields 4. The system validates the changes 5. The system commits the changes to its database 6. A confirmation message is displayed
<u>Post conditions:</u> • The updated user information is stored in the relational database
<u>Alternate flows:</u> • Invalid input: If invalid data is input, an error message is displayed, and the user is prompted to try again • Database error: If the update fails, the system logs the error, and the user is prompted to try again

Diagram 5.2.3: Login and basket interaction flow

Use Case: Basket management and checkout
Goal: To allow the user to select products, manage their basket, and complete a purchase
Actor(s): User
Preconditions: • The user is on the products page • The website is connected to a payment system
Main Flow: 1. The user selects one or more products to add to their basket 2. The system stores these products in an array that acts as the basket 3. The user selects their basket and is sent to the basket page 4. The user is prompted to update the basket as they desire then proceed to checkout 5. The system shows an order summary and prompts the user to enter payment and delivery details 6. Once the user has input their details, the system validates and processes the payment through a secure payment gateway 7. The system confirms the order and displays a success message or receipt
Post conditions: • The user's payment and delivery details are safely stored (if allowed to) • The basket is cleared •
Alternate flows: • Invalid payment: If payment validation fails, the user is prompted to try again • Session Timeout: If the session expires before checkout, the user has to restart the payment process

5.3 Interface Wireframes

The low-fidelity homepage wireframe is made using Figma, focuses on layout and structure. As mentioned in 4.3, it shows the placement of elements such as the navigation bar, hero banner, product grid, and header. Each section is clearly annotated to explain its purpose.

The high-fidelity wireframe, also created in Figma, represents the final visual design of the homepage and shows the vibrant colour palette that reflects PixelPals' brand identity.

Diagram 5.3.1: Low-fidelity homepage wireframe

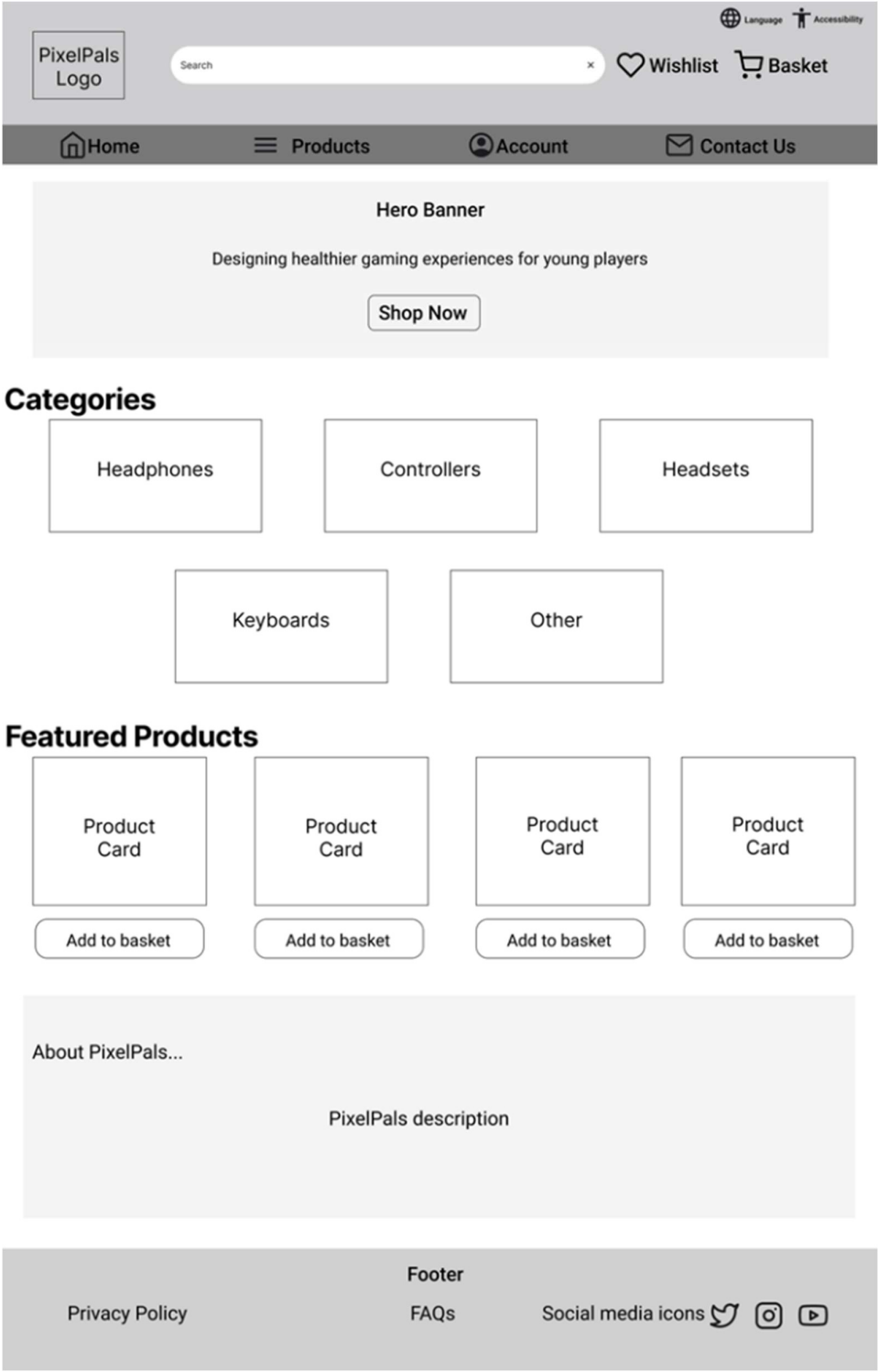
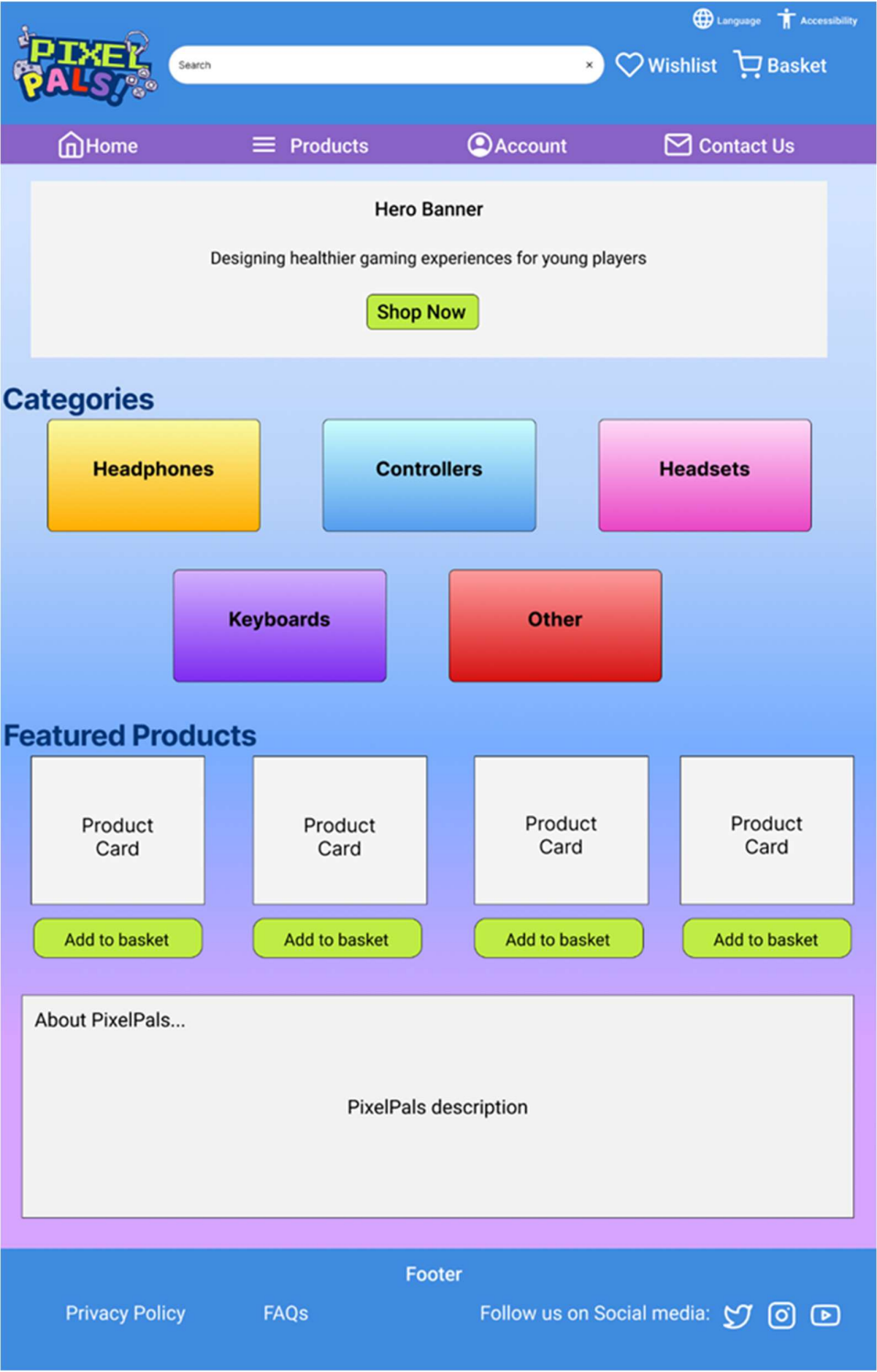


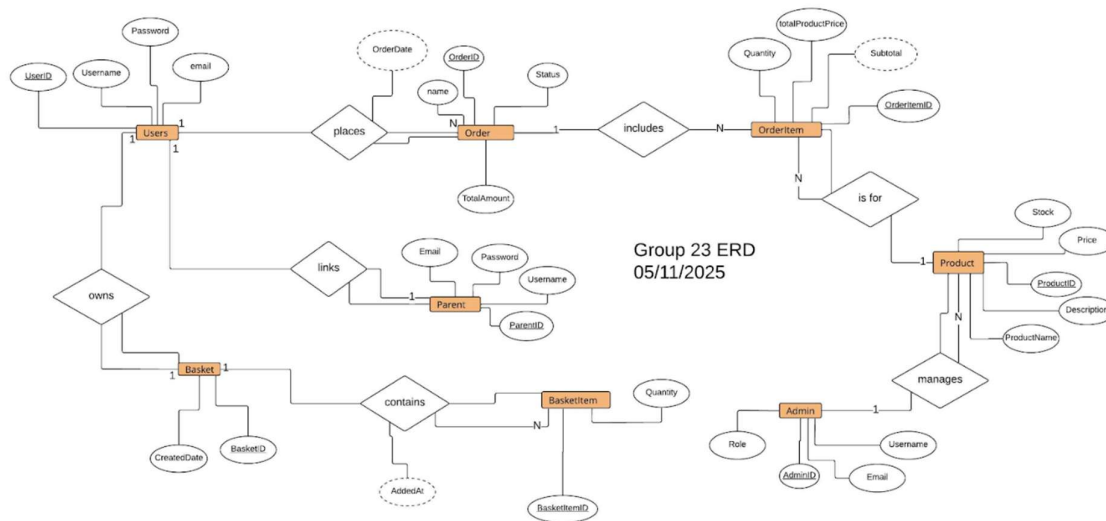
Diagram 5.3.2: High-fidelity homepage wireframe



5.4 Database Design (ERD)

Diagram 5.4.1: showing ERD

This ERD shows how users, orders, products, and related entities connect within the system. Entities are shown as rectangles, relationships as diamonds, and attributes as ovals—with underlined primary keys and dashed relationship attributes. Users can make multiple orders containing items linked to admin-managed products. Each user also has a basket with tracked items, and parents are linked for account management. The ERD maintains a clear, consistent, and efficient database structure.



5.5 Flow Diagrams (Admin/User Journey)

This user flow diagram shows the path going from landing on the homepage to then browsing products, adding the products to the basket and then checking out. The admin flow diagram shows the administrator actions, which include logging in, managing products and reviewing the orders.

Diagram 5.5.1: User journey flow through the main website pages.

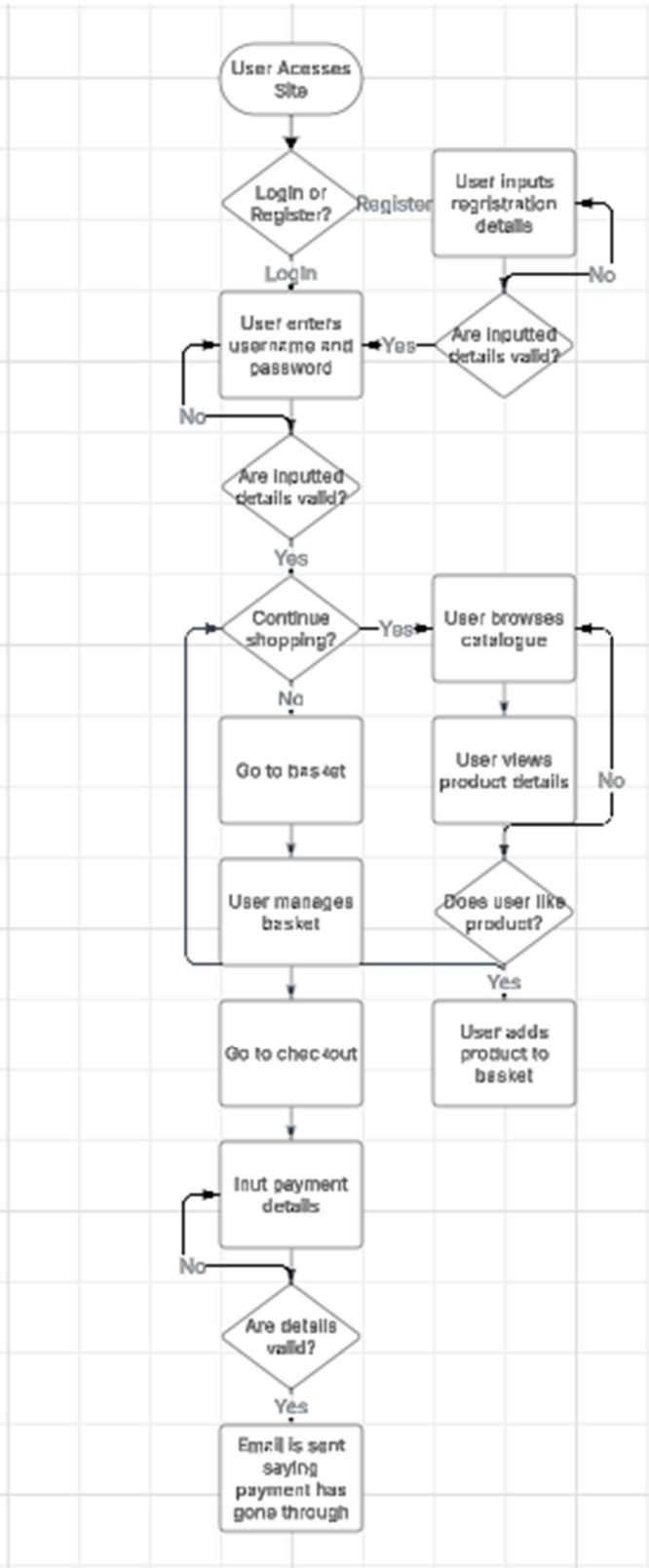
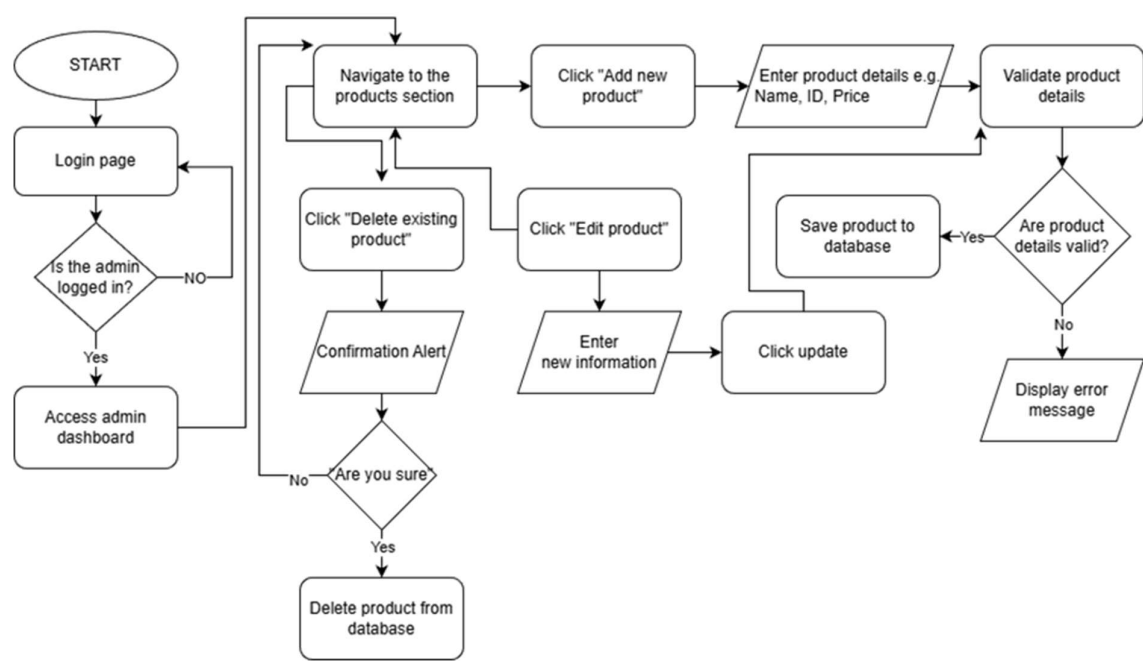


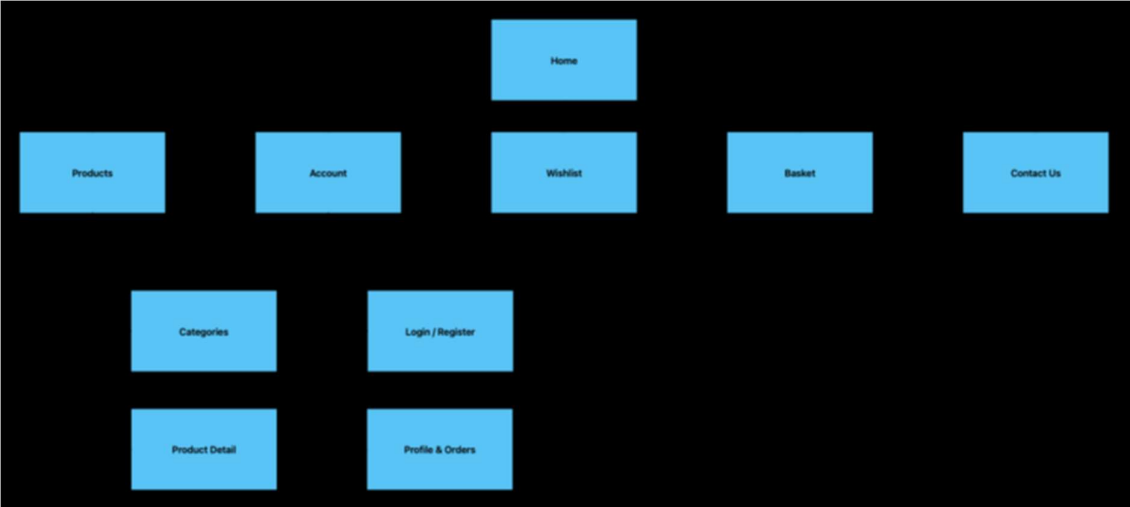
Diagram 5.5.2: Admin flow for managing products and orders.



5.6 Site Map

The site map visualises the overall structure of the PixelPals website, including the homepage, product categories, login and account pages, and any parent-specific sections. It ensures that navigation paths are clear and that key pages are easy to access.

Diagram 5.6.1: showing a simple site map



6. Project Management

6.1 Development Methodology

The PixelPals team adopted Agile methodologies to manage the development of the software requirements and design (SRD) report and to plan the structure of the PixelPals prototype.

The workload was structured into weekly sprints, with deliverables including drafted report sections from each team member and progress meeting notes. This approach targets issues specifically to ensure accuracy and efficiency, and allows for easy traceability.

Each week consisted of a weekly progress meeting, a plan for next week's progress meeting, individual task assignments via Trello, and updates on GitHub. The project manager ensured deadlines were realistic and that the progress aligned with the mark schemes. Team members were encouraged to communicate transparently and document each change for accountability and version control.

The cycle of planning, developing, reviewing and integrating mirrors Agile workflows in industry, and the team chose this method since it allowed adaptability and continuous improvement, which is important in software development.

6.2 Project Planning Tools (Trello)

The PixelPals team used Trello to plan, track and manage tasks for the Software requirements and design phase. The structure of Trello was visual, which kept the team time aware and organised.

The Trello Board was made up of lists, task cards and meeting cards. Each card was colour-coded by role, had due dates and checklists, and was moved across lists depending on its progress status. Table 6.2.1 explains the purpose of each list in Trello.

Table 6.2.1: Trello Lists and Their Purpose

List name	Purpose/Description
Week 6 to do	Contained all tasks planned for Week 6, such as setting up Trello, writing the SRD report draft structure, and organising meetings
Week 7 to do	Contained all tasks planned for Week 7, focused on completing SRD report sections and reviewing feedback from earlier drafts.
Week 8 to do	Contained all tasks planned for Week 8, final editing, formatting, and merging of the SRD report and Design alliance document before submission.
Currently doing	Showed tasks that were in progress or being reviewed by the project manager.
Upload to GitHub	Used to mark tasks and files that were ready to be uploaded to GitHub for version tracking and submission evidence.
Task card completed	Contained all finished tasks that were verified by the project manager or relevant lead.
Meetings done	Stored cards summarising weekly meetings and follow-up actions.
Documents/Templates/Rules	Contained shared files like the SRD report template, colour-code rules, and team guidelines to maintain consistency.

A link to the full Trello board is provided below for reference:

<https://trello.com/b/MgNq9DBY>

Trello board screenshots are stored in the GitHub repository under “Evidence/ Trello Screenshots”, and the full Trello layout showing all team roles and weekly progress is displayed in Appendix G (Figure G1).

6.3 Communication and Collaboration

The team communicated through Discord and in-person meetings. Emails were used as a formal reminder of deadlines, tasks, and meetings. Also, the team is divided into sub-groups that collaborate and overlap in their work. This combination assured consistent collaboration, availability, and understanding of task completion.

6.4 Version Control and File Management (GitHub)

The team used GitHub to store report drafts, evidence of project commitment and to track updates and changes. In terms of the process, each team member would make their own version of the branch and create commits of what was added or changed. When the update was finished, a pull request was sent to the project manager to review before merging it into the main branch.

The repository was named “Team 23 PixelPals” and was organised into clear folders:

- **Term1A/**: To keep GitHub submissions organised by term. It contains the Software Requirements Design (SRD) report, Docs and Evidence folders.
- **SRD report/**: SRD drafts from each member, submission plan, and final version of the SRD report.
- **Design alliance/**: Contains the Design Alliance document
- **Evidence/**: Trello screenshots, meeting notes, and commit summaries

This structure kept the project organised and traceable. Notes on commits and pull requests are stored in “Evidence/ GitHub commits” within the repository.

The GitHub repository link is provided below for reference:

https://github.com/maeesa-ali/team_23_pixelpals

6.5 Design and Technical Overview

The design of PixelPals focuses on creating child-centred shopping experiences that also appeal to parents. Saturated colours like Blue and Purple complement the logo. Lime green highlights key interactive elements, such as buttons like “Add to Basket” and “Shop Now,” encouraging engagement from users. The category tiles complement the logo with gradients to give a dynamic feel. Figma was used to design the wireframes, and Canva was used to create the logo and visuals.

The website will be developed using Laravel and MySQL to handle both the front-end layout and the server-side logic. PHP makes the website dynamic, for product listings and user interactions to be managed efficiently. The structure in the Figma wireframes will be implemented using responsive web design principles, ensuring compatibility across various screen sizes and devices.

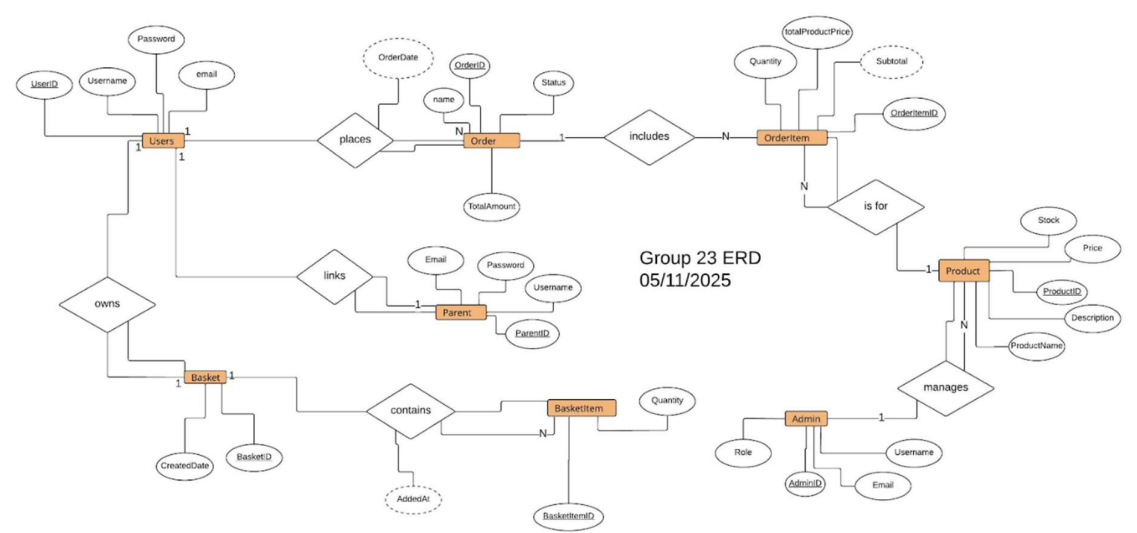
7. Conclusion

The development of the PixelPals Software Requirements and Design (SRD) document shows the team's ability to plan, collaborate and design a child-centred platform. Using

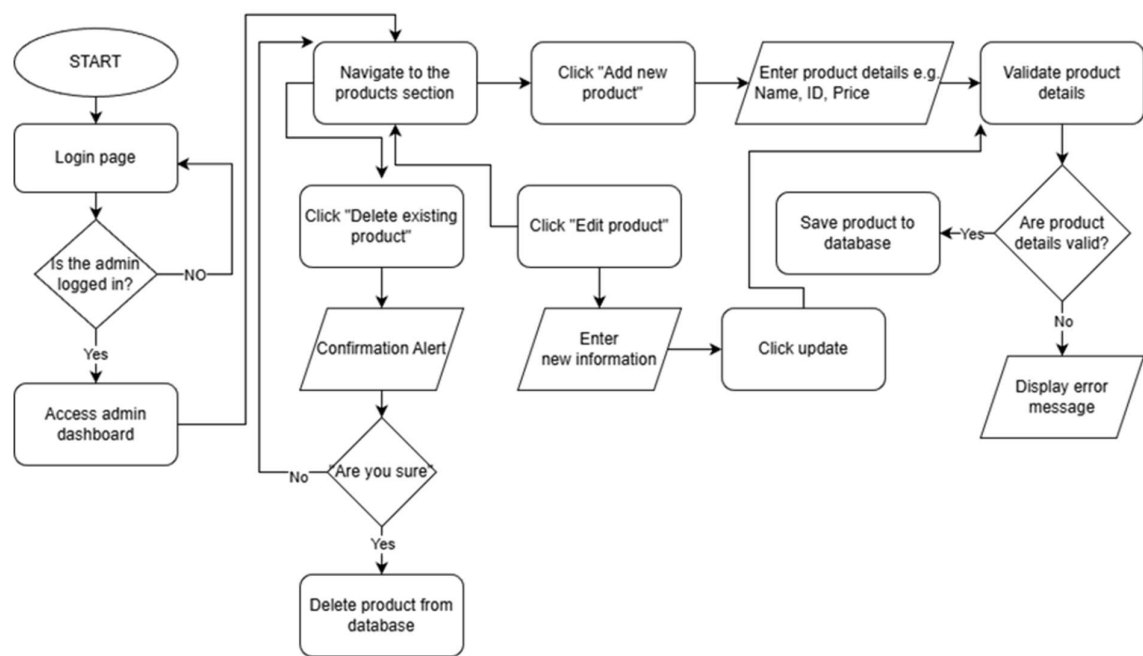
programming methods such as Agile and tools such as Trello, GitHub and Figma, whilst maintaining communication, has allowed the Team to progress efficiently through the project. The report provides clear instructions for future iterations using functional and non-functional requirements. Overall, PixelPals successfully combines creativity, technical planning and user-focused design to support healthy digital habits for children.

Appendices

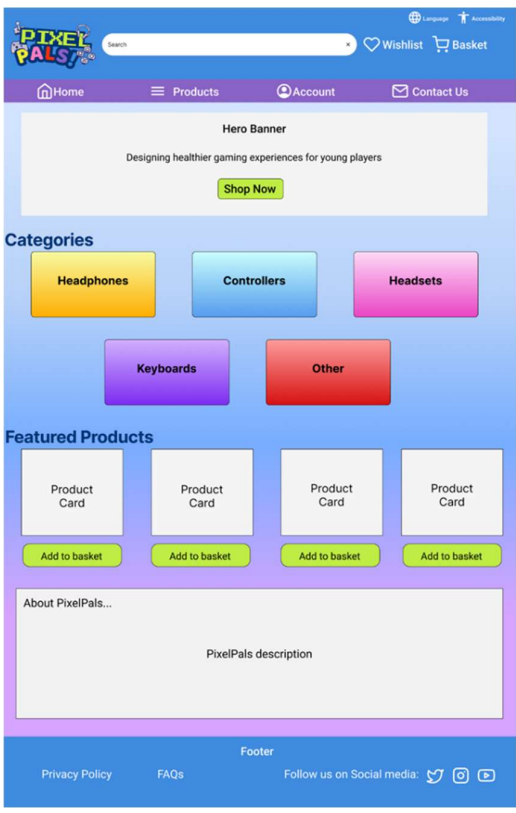
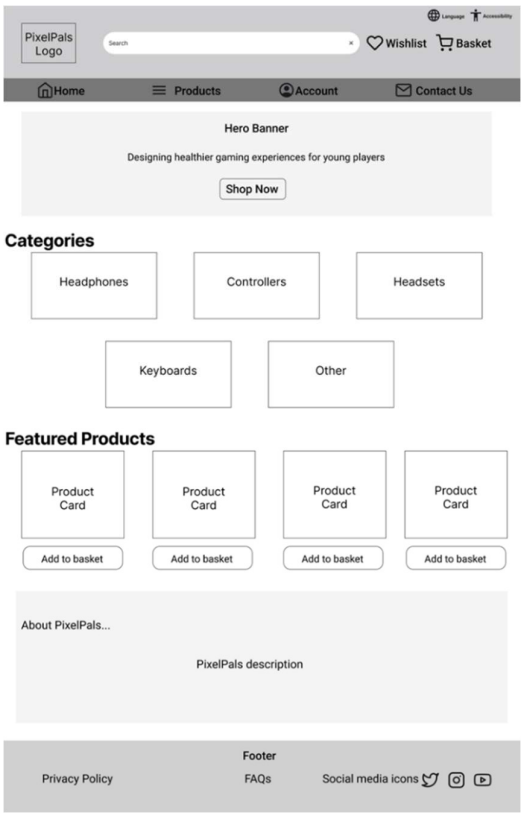
Appendix A – Entity Relationship Diagram



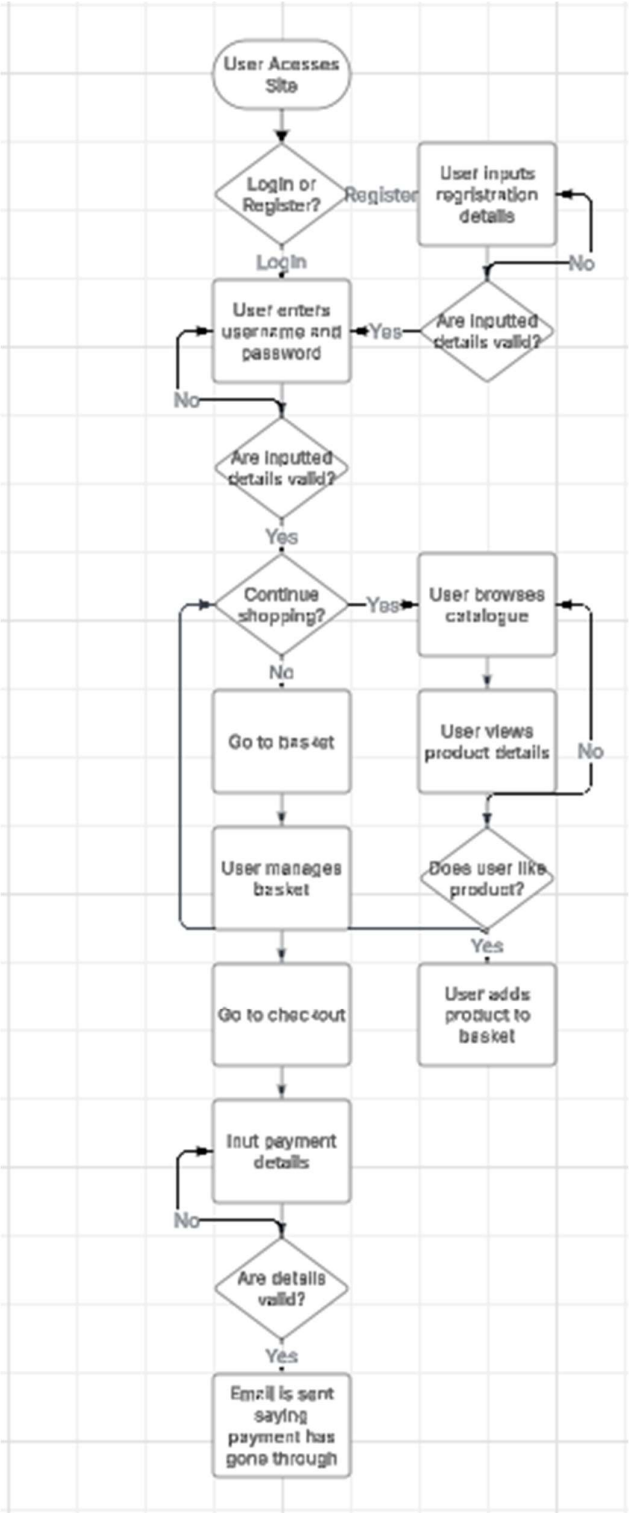
Appendix B – Admin Flow Diagram



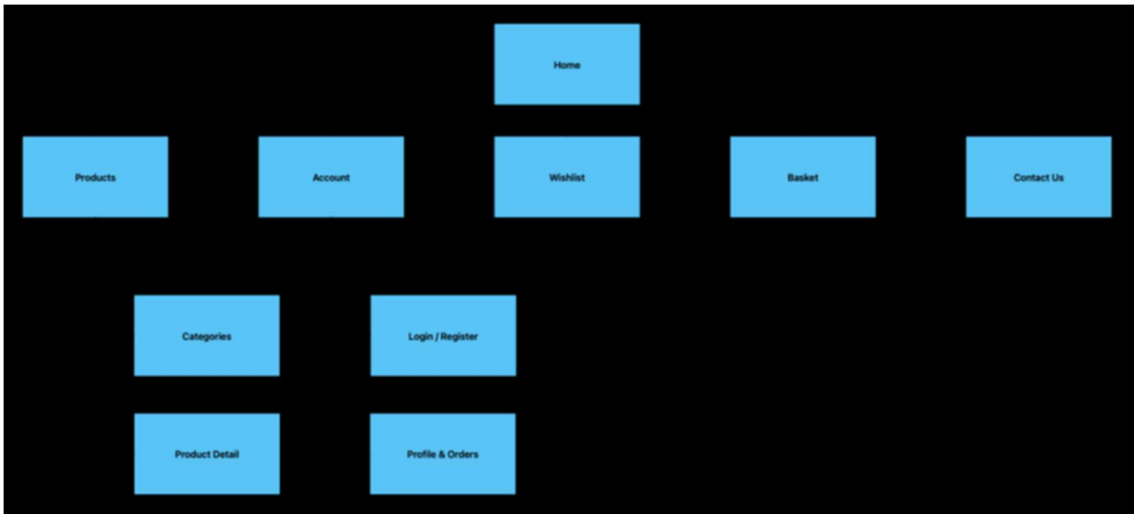
Appendix C – Wireframes and Brand Design



Appendix D – User Journey Diagram



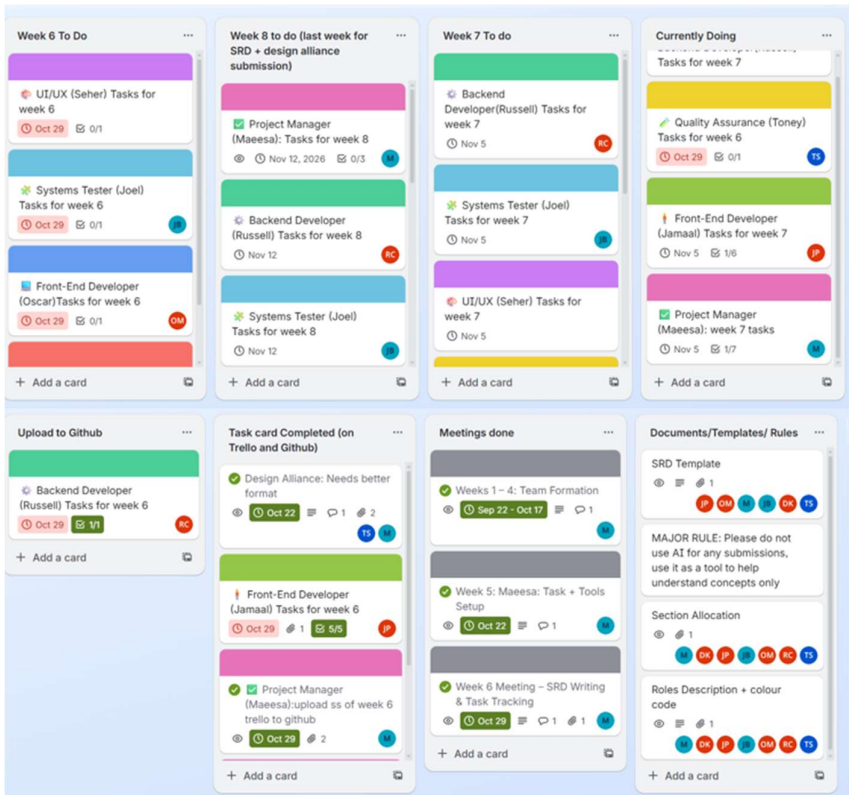
Appendix E – Site Map



Appendix F – Trello Board Structure and Workflow

Figure G1: PixelPals Trello Board Overview (Weeks 6–8)

This figure shows the team’s Trello board used to plan and track SRD report progress. Each list organised weekly tasks, deadlines, and responsibilities, and colour-coded cards displayed each member’s role and task status, supporting collaboration and clear time management.



8. References

Elghomati, A., Korhan, O., Sekeroglu, B. & Dogruiyol, K. (2022) 'Short-Term Impacts of Mobile Touch-Screen Device Use on Musculoskeletal Disorders During Covid-19 Pandemic: Risk Assessment Modelling and Verification', *South African Journal of Industrial Engineering*, 33(2), pp. 62-77. doi: 10.7166/33-2-2598.